

Group Assignment I

Advanced Econometrics 2026–2027

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Question 1

First write out $\sigma_{t+1}^2(\sigma_t^2, \epsilon_t)$, i.e. write the volatility equation as a function of the innovations.

We know:

$$x_t = \mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t$$

and by the given equation for σ_t we know:

$$\sigma_{t+1}^2 = \omega + \left(\alpha + \delta \tanh(-\gamma x_t) \right) \left(\frac{x_t - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2.$$

Substituting now gives us:

$$\begin{aligned} \sigma_{t+1}^2 &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \left(\frac{(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \left(\frac{\sigma_t\epsilon_t}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \epsilon_t^2 + \beta\sigma_t^2. \end{aligned}$$

We have now written $\sigma_{t+1}^2(\sigma_t, \epsilon_t) = \omega + \left[\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right] \epsilon_t^2 + \beta\sigma_t^2$. We want to show that $\{\sigma_t\}_{t \in \mathbb{Z}}$ is SE. We can do this by Bougerol's Theorem.

A1 $\{\epsilon_t\}_{t \in \mathbb{Z}}$ is SE. This holds since $\{\epsilon_t\} \sim TID(v)$ is iid and any iid sequence is SE.

A2 There should exist a finite initialization σ_1^2 such that $\mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] < \infty$. We have:

$$\begin{aligned} \mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] &= \mathbb{E}\left[\log^+ \left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \\ &\leq \mathbb{E}\left[\left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \quad (\log^+ |z| \leq |z|) \\ &\leq |\omega| + \mathbb{E}\left[\left| \alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1)) \right| \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}\left[(|\alpha| + |\delta| |\tanh(\dots)|) \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}\left[(|\alpha| + |\delta|) \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (-1 \leq \tanh(\cdot) \leq 1) \\ &= |\omega| + (|\alpha| + |\delta|) \mathbb{E}[\epsilon_1^2] + |\beta| |\sigma_1^2| \\ &< \infty. \end{aligned}$$

This is finite since $\omega, \alpha, \beta, \delta$ are finite parameters, $\mathbb{E}[\epsilon_1^2] < \infty$ for $\epsilon_1 \sim TID(v)$ with $v > 2$, and σ_1^2 is a finite initialization. So A2 holds.

A3 The contraction holds when

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \epsilon_t)}{\partial x} \right| \right] < 0.$$

Here $\phi(\sigma_t^2, \epsilon_t) = \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t))] \epsilon_t^2 + \beta\sigma_t^2$.

We recall also that $\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x)$.

By the chain rule,

$$\frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} = \delta \varepsilon_t^2 \left(1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right) \cdot (-\gamma) \left(\lambda + \frac{\varepsilon_t}{2\sqrt{\sigma_t^2}} \right) + \beta.$$

We then bound,

$$\begin{aligned} \left| \frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} \right| &\leq |\gamma| |\delta| \varepsilon_t^2 \left| 1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right| \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (\text{triangle inequality}) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (0 \leq 1 - \tanh^2(\cdot) \leq 1) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\omega}} \right) + |\beta| \quad (\sigma_t^2 \geq \omega \Rightarrow \frac{1}{2\sqrt{\sigma_t^2}} \leq \frac{1}{2\sqrt{\omega}}) \\ &= |\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta|, \end{aligned}$$

where $\sigma_t^2 \geq \omega$ holds for all t since $\alpha > |\delta|$ and $|\tanh(\cdot)| \leq 1$, $\alpha + \delta \tanh(-\gamma x_{t-1}) \geq \alpha - |\delta| > 0$, so the middle term of the σ_t^2 -recursion is non-negative and since $\beta \geq 0$ so is $\beta \sigma_{t-1}^2$. Hence $\sigma_t^2 \geq \omega$.

Therefore,

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \varepsilon_t)}{\partial x} \right| \right] \leq \mathbb{E} \left[\log \left(|\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta| \right) \right] < 0,$$

which is exactly the condition given in the question.

Thus, by Bougerol's theorem, $\{\sigma_t^2\}_{t \in \mathbb{Z}}$ converges to a SE process. Since $x_t = \mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t$ is a measurable function of the jointly SE sequence $\{(\sigma_t, \varepsilon_t)\}_{t \in \mathbb{Z}}$, in this case $\{x_t\}_{t \in \mathbb{Z}}$ is also SE by Krengel's theorem.

Question 2

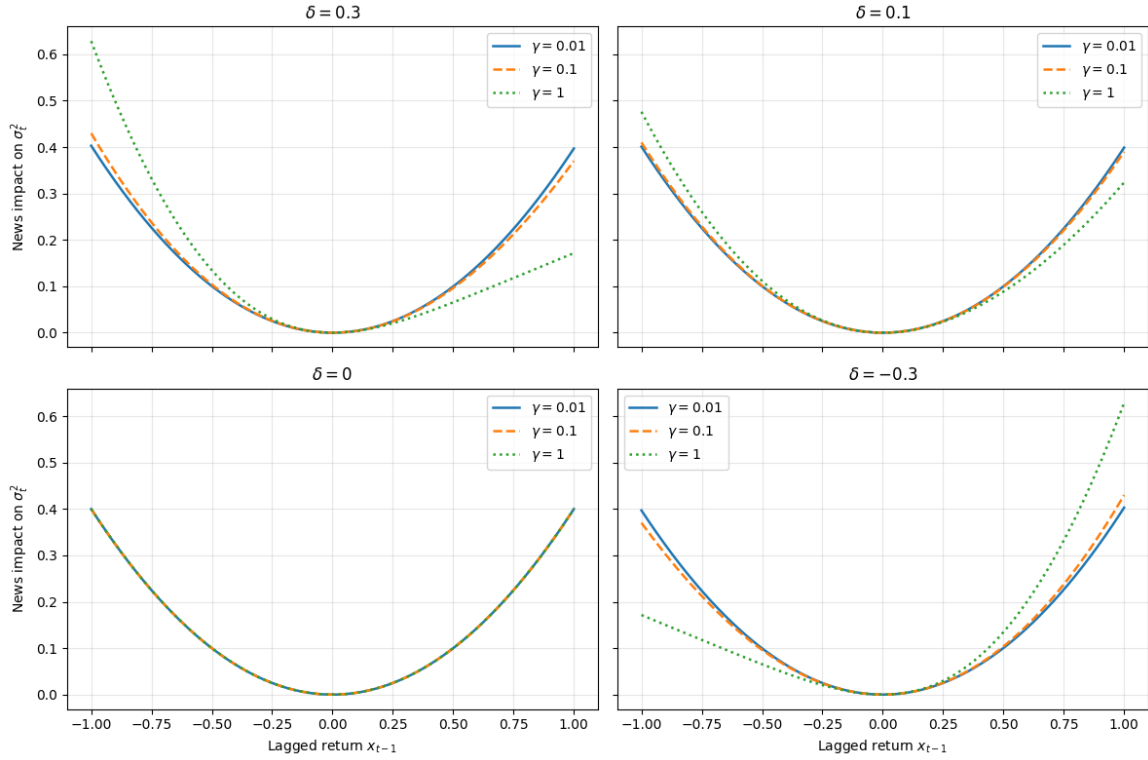


Figure 1: News impact curves of the GARCH-M-L model ($\mu = \lambda = 0$, $\alpha = 0.4$, $\sigma_{t-1}^2 = 1$)

Question 3

Table 1: Descriptive statistics of the daily holding period returns

	PFE	JNJ	MRK	AAPL
Number of observations	3270	3270	3270	3270
Mean	0.041	0.046	0.057	0.107
Median	0.000	0.036	0.046	0.089
Standard deviation	1.368	1.076	1.306	1.783
Minimum	-7.735	-10.038	-9.863	-12.865
Maximum	10.855	7.998	10.408	11.981
Skewness	0.223	-0.150	0.083	-0.076
Excess kurtosis	5.294	9.370	6.637	5.360

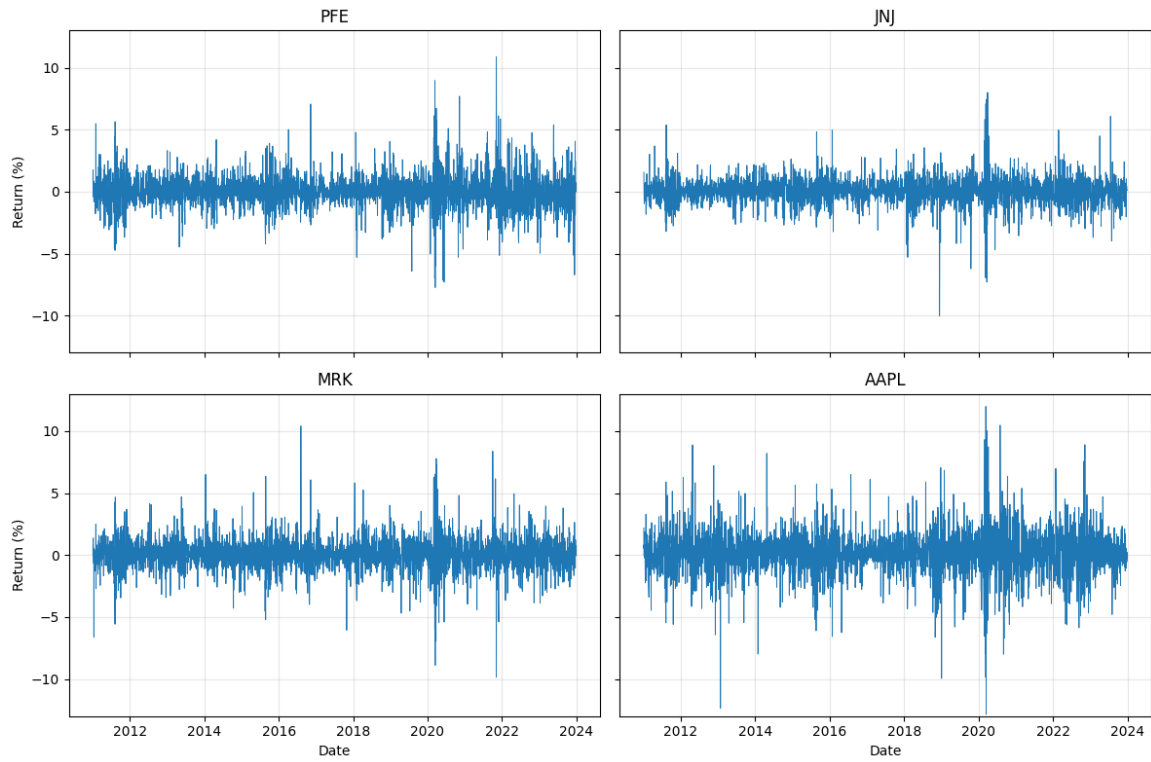


Figure 2: Daily holding period returns of Pfizer, Johnson & Johnson, Merck & Co., and Apple.

Question 4

Table 2: Maximum likelihood estimates of the three volatility models

	PFE			JNJ			MRK			AAPL		
	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3
μ	0.037	-0.028	-0.031	0.072	0.058	0.033	0.066	0.077	-0.046	0.154	0.072	0.108
λ		0.111	0.104		0.031	0.063		-0.015	0.117		0.061	0.022
ω	-0.027	-0.028	-0.025	-0.002	-0.002	0.003	0.014	0.015	0.019	0.038	0.037	0.012
α	0.036	0.036	0.043	0.025	0.025	0.026	0.039	0.040	0.040	0.090	0.089	0.073
β	0.961	0.963	0.953	0.923	0.924	0.916	0.906	0.904	0.906	0.874	0.875	0.915
δ			0.029			0.017			0.039			0.071
γ			0.410			1.228			4.742			0.439
ν	4.651	4.677	4.903	4.394	4.389	4.556	4.498	4.501	4.715	4.146	4.138	4.401
Log-lik	-3769	-3767	-3753	-3283	-3283	-3270	-3893	-3893	-3868	-4662	-4662	-4632
AIC	7549	7546	7522	6576	6577	6557	7796	7798	7752	9335	9335	9280
BIC	7578	7581	7569	6605	6612	6603	7825	7833	7799	9364	9370	9327

Notes: M1, M2, and M3 are denoted as the GARCH, GARCH-M, and GARCH-M-L models, respectively.

Question 5

Question 6